AUGSBURG UNIVERSITY

CSC 373 Artificial Intelligence, Spring 2026, 4 credits

This course explores the theoretical foundations of modern Artificial Intelligence. The process of using AI to solve a problem is one of choosing an appropriate technique, framing the problem for that context, implementing the technique (typically fine tuning parameters), and assessing the quality of that solution. As a student in the course, you will learn about categories of problems that can be addressed by AI, and you will learn a wide range of AI techniques drawn from myriad disciplines, such as graph theory, statistics, mathematics, formal logic, and neurobiology.

We begin the course by defining search in the context of a solution space, which is fundamental to many AI techniques. It becomes clear, very quickly, that simple search techniques are no match for the massive search spaces of complex problems. This leads to more intelligent, directed search techniques that use problem knowledge and/or random search, such as A*, minmax with alpha-beta pruning in adversarial search, constraint satisfaction, simulated annealing, and genetic algorithms. You will also learn about formal logic, which offers an alternative problem framework that employs inference to solve single goal state and planning problems. Finally, we will consider machine learning techniques that employ search through supervised, unsupervised, and reinforcement learning. You will learn how regression, Artificial Neural Networks, and Q-learning can be used for pattern classification, prediction, planning, and closed-loop control for a logical agent acting in the world.

Essential Course Information

LECTURE:
TuTh 2:00 pm - 3:30 pm in Hagfors 367
(Advanced Lab)

Instructor: Dr. Amy Larson
Pronouns: she/her/hers
Office: Hagfors 369C
Email: larsonam@augsburg.edu

Required Reading Material: "Artificial Intelligence: Foundations of Computational Agents" by David L. Poole and Alan K. Mackworth. Available free on-line at https://artint.info/3e/html/ArtInt3e.html or for purchase. Also: Introduction to Artificial Intelligence

Recommended Reading: "Artificial Intelligence: A Modern Approach" by Russell and Norvig. This textbook has been the de facto text for AI for decades. It is not required for this course because of its cost. Dr. Larson will have multiple copies available for checkout. There is supplemental material available through the website: https://aima.cs.berkeley.edu/

Required Applications and Technology: Personal laptop or desktop, Python, git, and a GitHub account.

Prerequisites: Receiving a P or C- or higher in CSC321 Software Design and Development, CSC351 Algorithms, MAT302 Discrete Mathematics, and MAT145 Calculus (or the equivalent of these). *If you have a grade of C+ or below in any prerequisite course, please budget additional time to work on this course.*

Drop-In (no appointment necessary) Student Hours are posted outside Dr. Larson's door in Hagfors 369C and on her Google calendar. To view, go to your Augsburg Google calendar and type *larsonam* in the *Search for people* box.

Student Learning Outcomes

Students can do the following at the end of the semester …

- Be able to list various AI techniques and the types of problems that they can solve.
- For a given AI technique, list how the heuristics, tuning parameters, and/or configuration impact the quality of the solution.
- Given a specific problem description, identify AI techniques that would be appropriate for solving the problem. Also, assess the advantages and disadvantages of those techniques in this context.
- Given a specific problem description and a Python implementation of an AI technique, formulate the problem so that it can be solved computationally with that technique. Also, assess the quality of the solution and discuss what contributes to its quality in this context.
- In Python, be able to implement a general framework for a given AI technique that can be applied to a variety of specific problems.
- Have an appreciation for the impact of AI on society and state ethical questions to be posed while using and advancing AI as a professional developer.

Student Background Knowledge

Students should be able to do the following before starting the semester ...

- Have basic mathematical knowledge including basic algebra, working with logarithms, solving for angles and sides of a triangle, graphing functions, and understanding derivatives.
- Analyze the runtime and space requirements of a given algorithm using big-O notation.
- State what an optimization problem is and assess the size of the solution space.
- Be able to implement basic graph algorithms, such as breadth-first search, depth-first search, and Dijkstra's single source shortest paths.
- Given Python code, be able to independently read through and understand how to use it. And for any aspect that you do not understand, be able to seek out appropriate resources to help you understand.
- Be able to construct a complex solution in Python that encompasses multiple files and modules. Furthermore, know how to find appropriate resources and repositories to build a complex solution.
- Have excellent technical and communication skills with respect to managing your own computer, installing packages, maintaining a GitHub repository, working in a team, documenting code, and explaining technical concepts to both non-technical and technical audiences.

How to Achieve Academic Success

Participation

You have to participate to be successful.

- Commit to the process of learning. Be persistent, dedicated, and engaged.
- Challenge yourself. This is hard work, but **you can do it**.
- Pay attention and participate in class. Ask questions. Be curious.
- Give yourself the time and space to work on this course outside of class time.
- Work early and often on your assignments.

Communication

Education is a team effort that requires significant participation of both student and instructor to be successful. Communication is a key component.

Communicate with Dr. Larson when …

- You need help understanding content or help with an assignment. You are not alone in this process. PLEASE do not be shy about asking for help.
- You need clarification on an assignment, policy, grade, etc.
- There is an ongoing situation that is interfering with your ability to be successful in the course.

To communicate with Dr. Larson, you might:

- Ask before, during, or after class.
- Attend drop-in student hours.
- Ask for an appointment to discuss grades or other personal matters.
- Send an email.

There are many ways that Dr. Larson will communicate with you, for example:

- During Class
- Moodle Site
- Announcements
- Assignment Directions
- Through Navigate
- Email

It is your responsibility to …

- Attend class and listen for information.
- Carefully read information and instructions provided in any and all forms.
- Check email and Moodle regularly (at a minimum, prior to every class period).
- Respond to communications when requested.

Please note that you should communicate with the CLASS office regarding accommodations.

If you miss 1 class, it is your responsibility to get notes from one of your fellow classmates. If you miss 2 consecutive classes, please check in with Dr. Larson as soon as possible. Given a TuTh schedule, every class missed puts you at risk of falling significantly behind.

Assessing Learning and Grades

How are the points distributed?

You will be graded on your participation in class and the weekly assignments that involve programming and written questions. This will build your portfolio of acquired knowledge and skills, which you will periodically review with Dr. Larson. The semester concludes with a programming project of your choosing using one of the techniques learned in class.

DESCRIPTION	PERCENTAGE
Weekly Participation	20%
Portfolio Review	65%
Final Project	15%
Total	100%

What assignments will I be doing?

In-class discussions and exercises will introduce you to the topics, give you an opportunity to practice applying the concepts, give you feedback on your understanding, and provide time for you to ask questions. Weekly homework in the form of written problems and programming assignments will help you gain a deeper understanding of that material and to build your portfolio. Your work should demonstrate university-level thoughtfulness and rigor.

Your weekly work will be gathered into a portfolio shared with Dr. Larson via Google Drive for written work and Gitlab for programming. You will present your portfolio to Dr. Larson every 3-4 weeks. During that portfolio review, you will engage in a discussion about your work and the concepts you have learned throughout the semester. You will receive a grade for each portfolio review.

Participation is assessed through a combination of your participation during class (not just attendance, but participation) and the quantity and quality of your work in your portfolio. You will receive a grade each week for participation.

In the final weeks of the semester, you will design and implement a project of your choosing that highlights a technique learned in class. For this project, you will state a hypothesis, implement an AI solution, run experiments to test your hypothesis, and arrive at a conclusion based on your experiments. You will present your work publicly either as a poster or a talk.

Participation and Portfolio Expectations

The approach to grading in this course is probably quite different from any other course that you have taken. This is an upper division course, so the expectation is that you are responsible for your learning and you are responsible for demonstrating your understanding of the material. Dr. Larson will look through your portfolio on a weekly basis to ensure that you are meeting expectations and that you will be sufficiently prepared for your portfolio review. However, your weekly work will not be graded in the traditional sense. We will discuss these assignments in class, which is information that you can use to assess your work, understanding, and progress in the course.

How do points correspond to letter grades?

CS GRADE SCALE	
Grade	**Percentage**
A	93-100
A-	90-92.99
B+	88-89.99
B	82-87.99
B-	80-81.99
C+	78-79.99
C	72-77.99
C-	70-71.99
D/LP	60-69.99
F/N	0-59.99
"P" is equivalent to C-/70 or higher.	

Grading Policies Related to Earning Credit Towards Your Degree

Disclaimer: All course policies subject to change with notice.

To receive a grade of P or C- or higher in this course, students must meet the following requirements:

- Earned an average grade of 60% or higher across all portfolio reviews.
- Earned a total of 70% or higher overall in the course.

Regardless of points earned, students may not earn more than 1 letter grade above their overall portfolio grade. For example, if a student earns 85% (a B) overall in the course, but the portfolio review grade is at 65% (a D), then the student will receive a C in the course.

Course Credit and Meeting Prerequisites

- For this course to count toward the major or minor in computer science, data science, or mathematics, you must earn a grade the equivalent of C- or higher grade. A grade of D, LP, or F will NOT count toward these degree requirements. The grading scale is listed above.
- If you take the course Pass/No Credit, you must earn the equivalent of a C- or higher to receive the grade of Pass.
- If you take the course Pass/No Credit, and you earn the equivalent of a D in the course, you will receive a Low Pass and earn credit for the course.

The instructor reserves the right to lower your grade if your attendance, participation, preparation, or behavior falls well below expectations. If during the term you are not progressing towards a passing grade or if your attendance, participation, preparation, or behavior is falling well below expectations, the instructor will typically send an Academic Alert.

Topics Schedule

WEEK	TOPIC	
1	Course Introduction. Uninformed Search	
2	Informed (Intelligent) Search	
3	Constraint Satisfaction and AC3	
4	Random Search (Genetic Algorithms and Simulated Annealing)	5-minute check-in
5	Search Reflection	Portfolio Review
6	Adversarial Search	
7	Logic, Inference, and Prolog	
8	Logic and Planning	
9	Spring Break	
10	Supervised Learning. Regression and Decision Trees	Portfolio Review
11	Supervised Learning: Feedforward Artificial NeuralNetworks	
12	Supervised Learning: Recurrent and Convolution ANNs	
13	Reinforcement Learning	
14	Large Language Models	
15	Projects	

Important University Dates

Description	Date
First Day of Instruction	Jan 20, 2026 (Tuesday)
Last day add/drop	Jan 07, 2026 (Wednesday)
Last day W at 50% refund	Feb 06, 2026 (Friday)
Last day to set grading option	May 19, 2026 (Tuesday)
Last day to withdraw**	Apr 10, 2026 (Friday)
Final grades due	May 12, 2026 (Tuesday)
Last Day of Instruction	May 01, 2026 (Friday)
Finals Week	May 04 to May 07

** Please consult with your academic advisor prior to dropping or withdrawing from the course to be sure that you understand the potential impact on your financial aid and academic progress, as well as to make alternative plans. It is also recommended that you discuss this with your instructor prior to withdrawing to go over your grade and alternatives to withdrawing.

Being Responsible and Accountable

All members of the Augsburg University community are expected to act with responsibility and moral integrity.

The student guide, including the conduct code, should guide your behavior. Academic honesty (Academic Honesty Policies) is integral to your success in this course and at Augsburg University. The following is an excerpt from the policy:

A University is a community of learners whose relationship relies on trust. Honesty is necessary for functioning of the Augsburg University community and dishonesty is, therefore, abhorred and prohibited. One example of how trust is destroyed by a particular form of dishonesty is found in plagiarism and its effects. In its 1990 'Statement of Plagiarism,' the American Association of University Professors (AAUP) Committee B on Professional Ethics notes that one form of academic dishonesty, plagiarism, 'is theft of a special kind [in which] a fraud is committed upon the audience that believes those ideas and words originated with the deceiver. Plagiarism is not limited to the academic community but has, perhaps, its most pernicious effect in that setting. It is the antithesis of the honest labor that characterizes true scholarship and without which mutual trust and respect among scholars is impossible.'

When you use someone else's work you must be sure you are not committing plagiarism or violating copyright. If you need assistance with proper citations, you may visit the Writing Lab, The Gage Center for Student Success, or the Lindell Library.

EXPECTATIONS for WORKING WITH OTHERS

- You help create a classroom environment that welcomes and values all students!
- You work cooperatively and respectfully with others in the class when asked to do so. You contribute to the work or discussion, and you allow others to contribute.
- You take responsibility for your part of any group assignment. This means that you are realistic about what you can contribute, you communicate in a timely manner when you are not meeting group expectations and deadlines, and you give credit to your teammates when they do more work than you.
- You offer reasonable assistance to a teammate who is communicating in a timely manner. This might include answering questions, coding together, or directing them to Dr. Larson.
- You share the responsibility of a group project, giving others an opportunity to contribute.

EXPECTATIONS for USING RESOURCES (including AI)

We are all learning how to best use AI as part of our curriculum. Policies and practices are evolving. Below are some standard policies, but policies might change during this semester (with notice). Let's keep talking about this and find the path that maximizes your learning!

- You may use claude.ai or chatGPT (or your favorite) to better understand concepts presented in class and in the textbook. You are welcome to use AI to learn new syntax or learn what an error message means. Use AI in the spirit of learning and cite your prompts.

- You are NOT permitted to use AI to write the answers to homework questions, to write code for assignments, or to aid with any other assignment that you submit for a grade, unless explicitly granted permission to do so. You can certainly understand that having someone or something else do your homework defeats the purpose of your education and circumvents learning!

- It is highly recommended that you do not use YouTube to learn the material. You are likely to waste hours watching videos with irrelevant content or teaching you a technique that is different from my expectations of how you should solve the problem.

- You must be able to verbally explain any answers or code that you submitted for grading. If you do not understand your answers or your code well enough to explain it when asked, you might receive a 0 on the assignment.

University Policies

Required Attendance Reporting Policy

Important Note Regarding Attendance
Federal law requires faculty to report students who never show up or who stop attending class. This report may cause a recalculation in a student's financial aid award, possibly requiring the student to return funds already received.

Non-Attendance/Non-Participation Defined:

Beginning of Term (student does not start): Non-Attendance/Non-Participation is defined as inactivity in the classroom setting in the **first 10 days** of the term and a failure to communicate with the instructor to explain that inactivity. Examples of inactivity may include: no in-person or on-line attendance, no online participation or Moodle activity, and/or a failure to complete any assignments.

End of Term (student starts but does not finish): Non-Attendance/Non-Participation is defined as a lack of activity in the classroom setting and a failure to communicate with the instructor to explain that period of inactivity. Examples of inactivity may include: no in-person or on-line attendance at class, an absence of Moodle activity, and/or a failure to complete assignments, exams, etc.

Health Policy

Health and Safety in the Classroom

In this class, we will follow all Augsburg University guidelines regarding contagious illnesses and individuals with compromised immune systems. All students are welcome to wear masks at any time, even if not requested by the university.

Please be respectful of the choices that the students around you are making. If the person next to you is wearing a mask, consider also wearing a mask. If you feel that someone is being unsafe with respect to your health, please feel free to talk to your professor about it.

Attendance

Be considerate of your community and do not attend class if you have COVID-19 or if you are experiencing symptoms consistent with any other spreadable illness. Students who miss class due to these conditions will not be penalized for their absence. *The same policy is in place for any contagious illness, for example covid-19, influenza, the stomach flu (rhinovirus or norovirus), strep throat, etc.*

As your instructor, I will trust your word when you say you are ill, and in turn, I expect that you will report the reason for your absences truthfully. This does not mean that you have to provide details, but a general category is helpful, such as a medical condition, transportation, family emergency, family obligation, work obligation, etc. This helps instructors better understand potential barriers to academic success that a student may be experiencing.

Disability Accommodations and Accessibility

Center for Learning and Adaptive Student Services (CLASS) at 612-330-1053, class@augsburg.edu, or stop by the Gage Center welcome desk.

All accommodations are arranged through the CLASS office. Please contact them as soon as possible to discuss potential accommodations.

> "If you have ADHD, a mental health disability, a learning disability, a physical/sensory impairment, a chronic health condition, are on the autism spectrum, or have another disability, the CLASS office can provide you with reasonable accommodations and support."

Augsburg University is committed to providing an environment where all students have the opportunity to equally participate in the academic experience, including students with disabilities. Students with disabilities have rights as determined by federal and state laws which require institutions to provide reasonable accommodations for the student's disability in order to afford an equal opportunity to participate in the university's programs, courses, and activities.

If you need accommodation because of a disability, please contact The Center for Learning and Adaptive Student Services (CLASS) at 612-330-1053, class@augsburg.edu, or stop by the Gage Center welcome desk on the link level of the Lindell Library to discuss how accommodations may be implemented in your Augsburg courses. If you have a letter from this office indicating you require academic accommodations, please be sure to check in with your instructor to ensure that you get the accommodations you need in this class.

Statement of Inclusion and Participation

Augsburg University values the diversity of persons, perspectives, and convictions. Critical thinking, rigorous analysis, and open discussion across a full range of ideas lie at the heart of the University's mission as an institution of higher learning. Essential to the University in living out its mission of educating students to be "informed citizens, thoughtful stewards, critical thinkers, and responsible leaders" is that the foundation be one of diversity, inclusion, equity, and intercultural competency.

We acknowledge that racism, classism, ableism, homophobia, transphobia, sexism, and other forms of oppression exist. In this class, people of all races, ethnicities, gender expressions and gender identities, religions, ages, sexual orientations, abilities, socioeconomic backgrounds, political affiliations, regions, and nationalities are strongly encouraged to share their rich array of perspectives and experiences. In doing so, we will be asked to take responsibility for our privileges and examine our words and actions. When we know better, we can do better.

Sexual Misconduct, Required Reporting, and Title IX

Augsburg University is committed to combating sexual misconduct. As a result, you should know that University faculty and staff members are required to report any instances of sexual misconduct, including harassment and sexual violence, to the University's Title IX Coordinator so that the student may be provided appropriate resources, support options and reporting options. What this means is that, as your professor, I am required to report any incidents of sexual misconduct that are directly reported to me, or of which I am somehow made aware. There are two important exceptions to this requirement:

- There are confidential resources in the Center for Wellness and Counseling (CWC) and Campus Ministry who do not have this reporting responsibility and can maintain confidentiality. More information can be found on page 15 of the Sexual Misconduct Policy: https://sites.augsburg.edu/studentaffairs/reporting/

- An important exception to the reporting requirement exists for academic work. Disclosures about sexual misconduct that are shared as part of an academic project, classroom discussion, or course assignment, are not required to be disclosed to the University's Title IX office.